Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Natalie Campbell	natcamp@udel.edu
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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is to create a web application that will generate a swim set for a workout

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are some websites or apps that already help swimmers create workouts, like MySwimPro, but most of these require subscriptions or go beyond a casual swim level.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My web application will allow swimmers to enter information such as skill level, workout length, and training focus. The program will then generate a swim set that matches the user's goals. The website is supposed to help swimmers create structured workouts without having to spend time planning them.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The goal of my program is to take user preferences as input, process them using predefined swim sets and rules, and generate a customized swim workout.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Skill level (beginner, intermediate, advanced) Practice length Total distance goal Workout focus (maybe previous times)

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will compare the user's input to a collection of workout sets and pick appropriate warmups, main sets, and cooldowns. It will then organize the set into a complete workout.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

A formatted swim workout with a warmup, main set, and cooldown.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Swimmer Workout Swim Set Exercise

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Need: - workout type - total distance - difficulty level - team Ignore: - pool location - time of day

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most challenging part will be creating a system that generates workouts that are balanced and appropriate for different skill levels. Another challenge will be making sure that the generated distances match the user's requested workout length.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I still haven't decided if I will be using an LLM but if I do I will make sure I read through the suggested code carefully, test it, and understand each function. I will compare suggestions with course materials and debug errors myself before asking for additional help.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal is to build a website that swimmers could realistically use to generate workouts. I want to improve my programming skills and become more comfortable using Drafter.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

"I think I want to create a web tool that could help generate swim workouts. Coming up with a workout set is the most annoying thing I have to do if I am trying to swim alone"

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

She said that she likes my idea. She also suggested that it could be helpful to add an input where the swimmer can put in previous times and have the program identify some strengths and weaknesses.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The most complex suggestion that my friend gave was having the program be able to store previous workouts and then adjust future workouts based on improvements. An alternative to having it store it was just to maybe add an extra input to allow spots for improvements so the program knows to avoid that specific part.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

After brainstorming, I narrowed my project to generating workouts based only on a few user inputs rather than building a complete training platform. This makes it more realistic and achievable.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use the textbook, lecture materials, office hours, TA's and maybe an LLM for clarifications.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

1. Read the full error message carefully
2. Identify which function or file caused the error
3. Test smaller sections of code separately
4. Compare my code with examples from class
5. Check drafter and bakery documentation
6. Use print statements to locate the problem
7. Ask LLM to explain
8. Visit office hours or ask a TA

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM for this assignment, I only used a partner to bounce ideas off of.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.